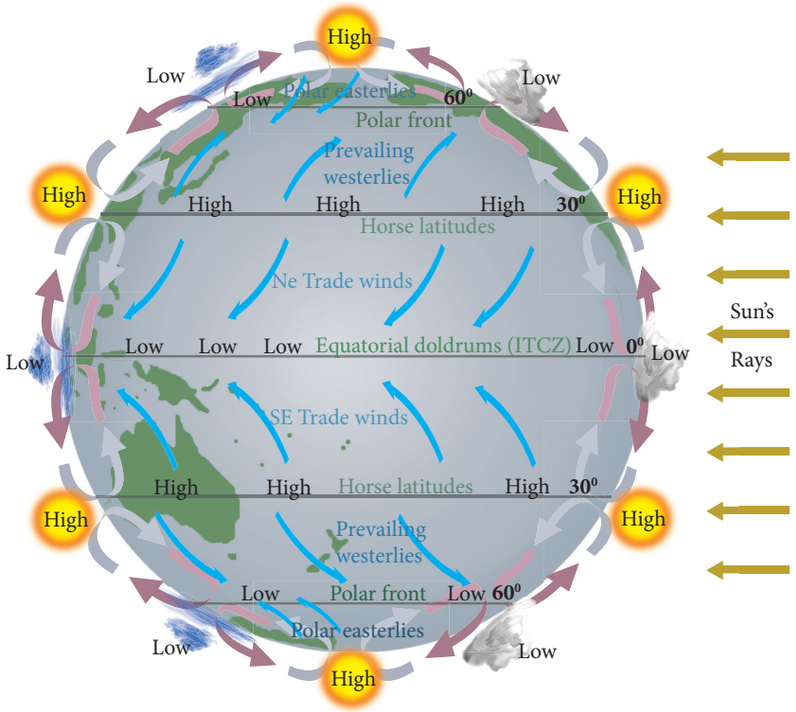


Figure 6.13 Pressure Belts and Primary Winds

6.4.6 Basis of Formation of Pressure Belts

There are two important bases on which the pressure belts are formed. They are;

a. Temperature: The equatorial low pressure and polar high pressure belts are formed due to high and low temperature respectively. So they are called as 'Thermally formed pressure belts'.



Ocean is dominant in the southern hemisphere between the latitudes 40°S and 60°S. Hence the westerlies are so powerful and persistent that the sailors used such expressions as **"Roaring Forties", "Furious Fifties"** and "**Screaming or Shrieking Sixties"** for these high velocity winds in the latitudes of 40°, 50° and 60° respectively.

b. Dynamism: The sub tropical high and sub polar low pressure belts are formed due to movement and collision of wind system. So they are called as 'Dynamically formed pressure belt system'.

6.4.7 Meridional Cell System

The cell along with trade winds, equatorial low and sub tropical high pressure belts is called as 'Hadley cell', meanwhile the cell formed by westerly wind along with sub tropical high and sub polar low pressure belt is called 'Ferrell's cell'. The cell at polar formed by polar easterlies with polar high and sub polar low pressure belt is called as 'Polar cell' (Figure 6.14).

ITCZ – Inter Tropical Convergent Zone

The region where both trade wind systems meet is known as 'Inter Tropical Convergent Zone'.

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